Modelling Software-based Systems

Proof Obligation Generation for Machines and Verification Machines

Telecom Nancy (IL et LE)

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Summary

1. CM3
2. Overview of machines, contexts and proof obligations
3. Proof Obligations for Contexts and Machines
   - PO thm/THM (context)
   - PO th/THM (machine)
   - PO evt/inv/INV
   - PO evt/act/FIS
4. Checking contracts using Event-B on the FIDE Rodin
   - Method for proving invariance properties
   - Verification of contract
   - Definition of machine for checking contract
   - Soundness of the technique
5. Example
6. Conclusion

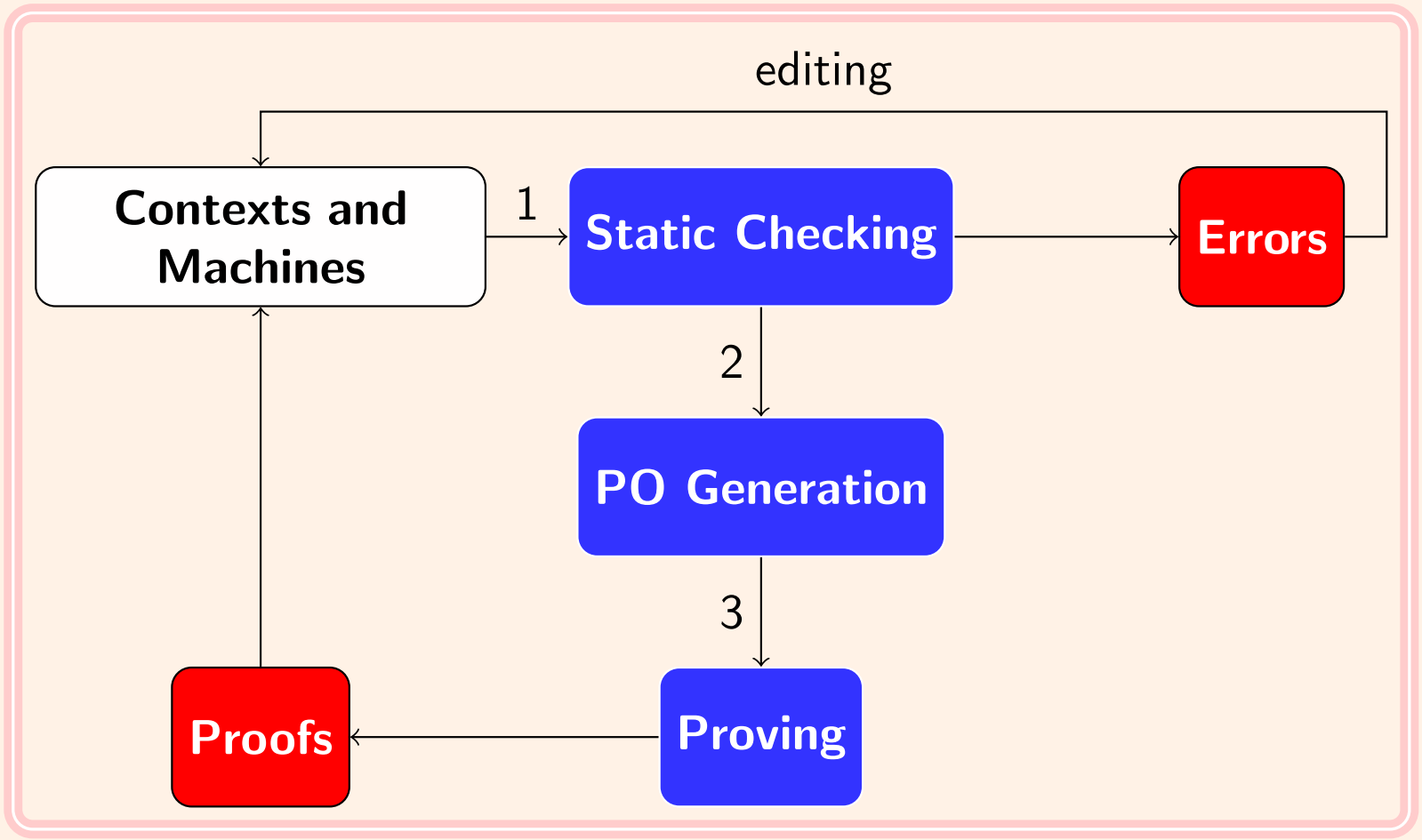
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Analysis of the Event-B Models



Machines en Event B

MACHINE
m
REFINES
am
SEES
c
VARIABLES
u
INVARIANTS
$I(s, c, u)$
THEOREMS
$Q(s, c, u)$
VARIANT
$exp(s, c, u)$
EVENTS
$INITIALIZATION$
$\ldots$
e
$\ldots$
END

Machines en Event B

MACHINE
m
REFINES
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$I(s, c, u)$
THEOREMS
$Q(s, c, u)$
VARIANT
$exp(s, c, u)$
EVENTS
$INITIALIZATION$
$\ldots$
e
$\ldots$
END

- $\Gamma(m)$: environment for the machine m defined by the context c and it provides a list of seen axioms $Ax(s, c)$ and a list of seen theorems $Th(s, c)$ for the sets s and constants c.
- $\Gamma(m) \vdash \forall u.\textsc{Init}(s, c, u) \Rightarrow \mathrm{I}(s, c, u)$
- For each event e in E :
  $\Gamma(m) \vdash \forall u, u'.\mathrm{I}(s, c, u) \wedge BA(e)(u, u') \Rightarrow \mathrm{I}(u')$
- For each event e in E :
  $\Gamma(m) \vdash \forall u.\mathrm{I}(s, c, u) \wedge GRD(e)(s, c, u) \Rightarrow \exists u'.BA(e)(u, u')$
- $\Gamma(m) \vdash \forall u.\mathrm{I}(s, c, u) \Rightarrow \mathrm{Q}(s, c, u)$
- Generated proof obligations are derived from those conditions.

Checking the well formation of Event-B expressions

- Event-B expressions are contexts, machines, properties, equations, set-theoretical expressions ...
- e is an Event-B expression and $\mathsf{wd}(e)$ is a logical property expressing the well definition of e.
- $\mathsf{wd}(1 = 2) \triangleq \mathsf{wd}(1) \wedge \mathsf{wd}(2)$
- $\mathsf{wd}(a/b) \triangleq b \neq 0 \wedge \mathsf{wd}(a) \wedge \mathsf{wd}(b)$
- $\mathsf{wd}(f(g)) \triangleq g \in dom(f) \wedge f \in A \nrightarrow B$

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PO th/THM

CONTEXTS
c
EXTENDS
ac
SETS
s
CONSTANTS
c
AXIOMS
$Ax(s, c)$
THEOREMS
$th_1 : P_1(s, c)$
$\ldots$
$th_n : P_n(s, c)$
$th : P(s, c)$
$\ldots$
END

s	*seen sets*
c	*seen constants*
$Ax(s, c)$	*seen axioms*
$Th(s, c)$	*previous proved theorems*
	$PTh(s, c) = \{P_i(s, c) \mid i\ 1..n\}$
$P(s, c)$	*property over s and c*

PO th/THM

$Ax(s, c), Th(s, c) \vdash P(s, c)$

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post

PO th/THM (machine)

MACHINE
m
...
VARIABLES
u
INVARIANTS
$I(s, c, u)$
THEOREMS
$Q(s, c, u)$
$th : P(s, c, u)$
...
END

s	*seen sets*
c	*seen constants*
u	*variables*
$Ax(s, c)$	*seen axioms*
$Th(s, c)$	*seen theorems*
$I(s, c, u)$	*invariants*
$Q(s, c, u)$	*theorems*
$P(s, c, u)$	*property over s,c and u*

PO th/THM

$$Ax(s, c), Th(s, c), I(s, c, u) \vdash P(s, c, u)$$

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PO evt/inv/INV

EVENT evt
  ANY x **WHERE**
    $G(x, s, c, u)$
  THEN
    $u :| BAP(x, s, c, u, u')$
  END

$BA(\text{evt}) \ \widehat{=}$

$$\exists x. \left(\begin{array}{l} \wedge\ G(x, s, c, u) \\ \wedge\ BAP(x, s, c, u, u') \end{array} \right)$$

$GRD(\text{evt}) \ \widehat{=}\ G(x, s, c, u)$

$ACT(\text{evt}) \ \widehat{=}\ BAP(x, s, c, u, u')$

s	*seen sets*
c	*seen constants*
u	*variables*
$Ax(s, c)$	*seen axioms*
$Th(s, c)$	*seen theorems*
$I(s, c, u)$	*invariants*
$Q(s, c, u)$	*theorems*
evt	*event name*
x	*event parameter*
$G(x, s, c, u)$	*event guard*
$BAP(x, s, c, u, u')$	*event before-after predicate*
$inv : inv(s, c, u')$	*specific modified invariant*

PO evt/inv/INV

$$Ax(s, c), Th(s, c), I(s, c, u), G(x, s, c, u), BAP(x, s, c, u, u') \vdash inv(s, c, u')$$

PO Q/THM $Ax(s, c), Th(s, c), I(s, c, u) \vdash Q(s, c, u)$

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PO evt/act/FIS

EVENT evt
  ANY x **WHERE**
    $G(x, s, c, u)$
  THEN
    $u :\mid BAP(x, s, c, u, u')$
  END

$BA(\text{evt}) \mathrel{\widehat{=}}$
$\left(\begin{array}{l} \wedge\ G(x, s, c, u) \\ \wedge\ BAP(x, s, c, u, u') \end{array} \right)$
$GRD(\text{evt}) \mathrel{\widehat{=}} G(x, s, c, u)$
$ACT(\text{evt}) \mathrel{\widehat{=}}$
$BAP(x, s, c, u, u')$

s	*seen sets*
c	*seen constants*
u	*variables*
$Ax(s, c)$	*seen axioms*
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PO evt/act/FIS

$Ax(s, c), Th(s, c), I(s, c, u), G(x, s, c, u), \vdash \exists u'.BAP(x, s, c, u, u')$

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Contract verification : what the technique involves

A program P *fulfils* a contract (pre,post) :

- P transforms a variable v starting from an initial value v_0 and producing a final value v_f : $v_0 \xrightarrow{P} v_f$
- v_0 satisfies pre : pre(v_0) and v_f satisfies post : post(v_0, v_f)
- pre$(x_0) \wedge x_0 \xrightarrow{P} x_f \Rightarrow$ post(x_0, x_f)

requires $pre(v_0)$
ensures $post(v_0, v_f)$
variables v

begin
$0 : P_0(v_0, v)$
$\texttt{instruction}_0$
...
$i : P_i(v_0, v)$
...
$\texttt{instruction}_{f-1}$
$f : P_f(v_0, v)$
end

- $pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$
- $pre(v_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$
- verification conditions for all pairs $\ell \longrightarrow \ell'$

Proving invariance and safety properties

Design of a method for proving invariance and safety properties for a programming language

- We consider a classical programming language denoted PROGRAMS
- and we assume that this programming language supports assignment, conditional statements, bounded iteration, unbounded iteration, simple or structured variables such as arrays, and the definition of constants.
- Let us consider a program P in PROGRAMS ; this program comprises
  - globally named variables v,
  - globally named constants pc,
  - types associated with the globally named variables VALUES and identified ẁith a set of possible values for the variables,
  - instructions following an order defined by the syntax of the programming language.

Hypotheses

- We define a set of control points $\textsc{Locations}$
- for each program or algorithm P. $\textsc{Locations}$ is a finite set of values and a hidden variable denoted pc traverses this set according to the sequence.
- the space of possible values $\textsc{Values}$ is a Cartesian product of the form $\textsc{Locations} \times \textsc{Memory}$
- the variables x of the system are decomposed into two independent entities $x = (pc, v)$ subject to the conditions $pc \in \textsc{Locations}$ and $v \in \textsc{Memory}$.

$$x = (pc, v) \wedge pc \in \textsc{Locations} \wedge v \in \textsc{Memory} \tag{1}$$

We consider an annotated program P ; we are given a relational model $\mathcal{MP} = (Th(s, c), x, \textsc{Values}, \textsc{Init}(x), \{r_0, \ldots, r_n\})$ where

- $Th(s, c)$ is a theory defining the sets, constants and static properties of this program
- x is a list of flexible variables, and x comprises a control part and a memory part.
- $\textsc{Locations} \times \textsc{Memory}$ is a set of possible values for x.
- $\{r_0, \ldots, r_n\}$ is a finite set of relations linking the values before x and the values after x', and conforming to the succession relation $\longrightarrow$ between control points.
- $\textsc{Init}(x)$ defines the set of initial values of (pc_0, v) and $x = (pc_0, v)$ with $pre(v)$ which characterises the initial values of v at the initial point.

Assumptions regarding the calculation

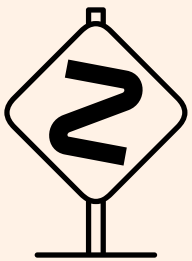
We assume that there exists a graph over the set of control values defining the flow relation, and we denote this structure $(\text{LOCATIONS}, \longrightarrow)$.

Definition

$\ell_1 \longrightarrow \ell_2 \stackrel{\Delta}{=} pc = \ell_1 \wedge pc' = \ell_2$

Definition

(Annotation of a control point)
Let $(\text{LOCATIONS}, \longrightarrow)$ be a structure and $\ell \in \text{LOCATIONS}$ a label ℓ. An annotation of a control point ℓ is a predicate $P_\ell(v)$ (assertion) or $P_\ell(v_0, v)$ (relational).



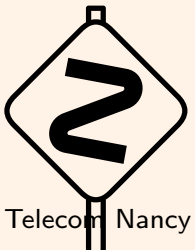
$P_\ell(v_0, v)$ expresses a relation between the initial value of V, denoted v_0, and v, the current value of V at point ℓ, and therefore $P_\ell(v_0, v) \Rightarrow pre(v_0)$ specifies that v_0 is an initial value.

Relationship between annotation and invariant

- The labels ℓ belong to $\textsc{Locations}$: $\ell \in \textsc{Locations}$.
- The variables v belong to $\textsc{Memory}$: $v \in \textsc{Memory}$.
- $pre(v_0)$ specifies the initial values of v.
- Whenever control is at ℓ, v satisfies $P_\ell(v)$: $pc = \ell \Rightarrow P_\ell(v_0, v)$.
- For any state (ℓ, v) of the program, the following property is true but must ĥe proved :

$$J(\ell_0, v_0, pc, v) \triangleq \left[\begin{array}{l} \wedge\ pc \in \textsc{Locations} \\ \wedge\ v \in \textsc{Memory} \\ \ldots \\ \wedge\ pc = \ell \Rightarrow P_\ell(v_0, v) \\ \ldots \end{array}\right.$$

- $J(\ell_0, v_0, pc, v)$ is an invariant constructed ỳrom the annotations generated, but we must show that this invariant enables us to verify the three conditions of the induction principle.



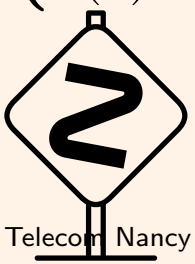
ℓ_0 denotes the label marking the start of the algorithm and ℓ_f is the end of the program. We may simply use 0 and f.

Checking to do

$$x = (pc, v) \text{ et } J(\ell_0, v_0, pc, v) \triangleq \left[\begin{array}{l} \wedge\ pc \in \textsc{Locations} \\ \wedge\ v \in \textsc{Memory} \\ \ldots \\ \wedge\ pc = \ell \Rightarrow P_\ell(v_0, v) \\ \ldots \end{array} \right.$$

Let $(Th(s,c), x, \textsc{Values}, \textsc{Init}(x), \{r_0, \ldots, r_n\})$ a relational model for a program. A property $\mathrm{A}(x_0, x)$ is a safety property for P if $\forall x_0, x \in \textsc{Locations} \times \textsc{Memory}. Init(x_0) \wedge x_0 \xrightarrow[\textsc{Next}]{} x \Rightarrow A(x)$.We know that this property implies that there exists a state property $\mathrm{I}(x_0, x)$ such that the three properties hold, but we apply this verification to J : $\forall x_0, x, x' \in \textsc{Locations} \times \textsc{Memory}$:

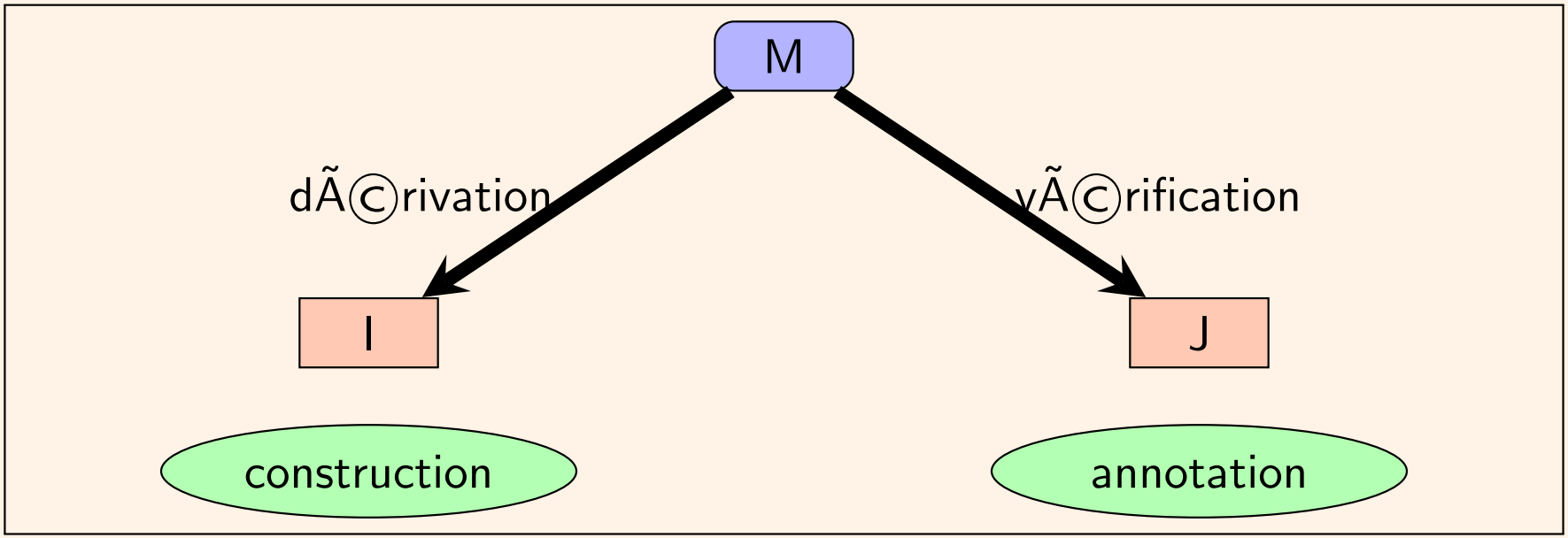
$$\left\{ \begin{array}{ll} (1) & \textsc{Init}(x_0) \Rightarrow \mathrm{J}(x_0, x_0) \\ (2) & \mathrm{J}(x_0, x) \Rightarrow \mathrm{A}(x_0, x) \\ (3) & \forall i \in \{0, \ldots, n\} : \mathrm{J}(x_0, x) \wedge x\ r_i\ x' \Rightarrow \mathrm{J}(x_0, x') \end{array} \right.$$



$\forall i \in \{0, \ldots, n\} : \mathrm{J}(x_0, x) \wedge x\ r_i\ x' \Rightarrow \mathrm{J}(x_0, x')$ is équivalent to $\mathrm{J}(x_0, x) \wedge (\exists i \in \{0, \ldots, n\} : x\ r_i\ x') \Rightarrow \mathrm{J}(x_0, x')$

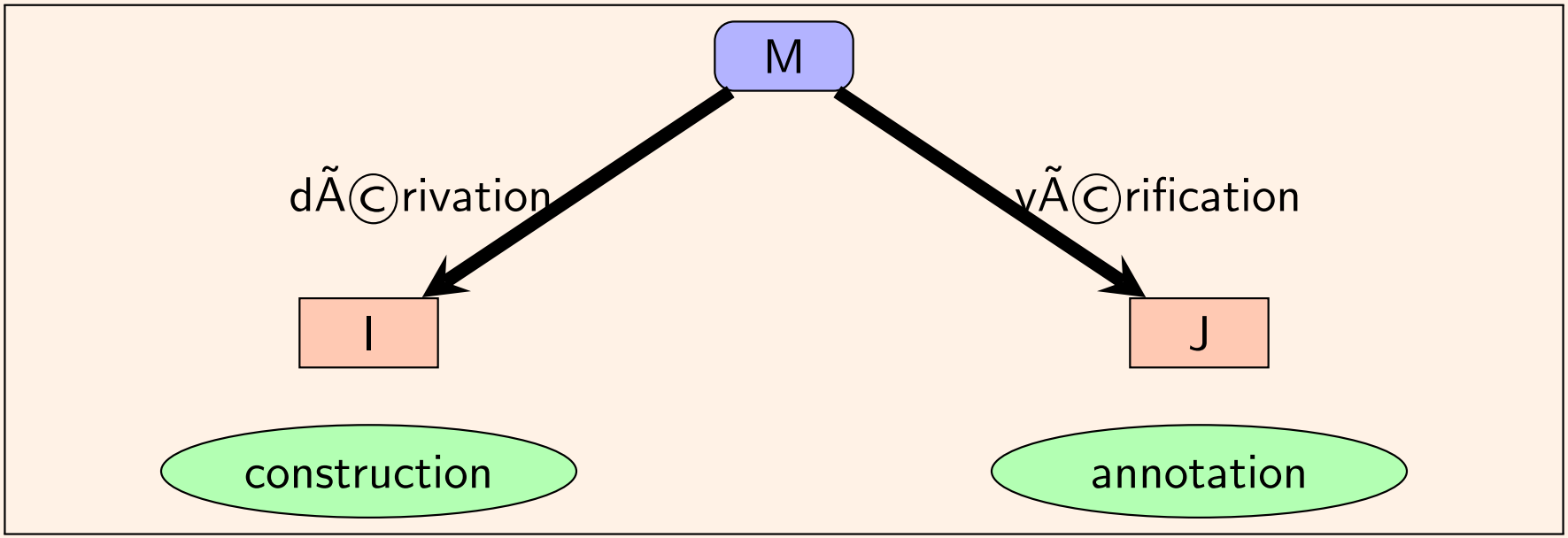
Checking to do

- Application of the correctness to the relational principle of induction : if we verify the three properties, then A is a safety property for the model in question (verification).
- If we wish to show that A is a safety property, then we must use the invariant to construct annotations for the model (derivation).



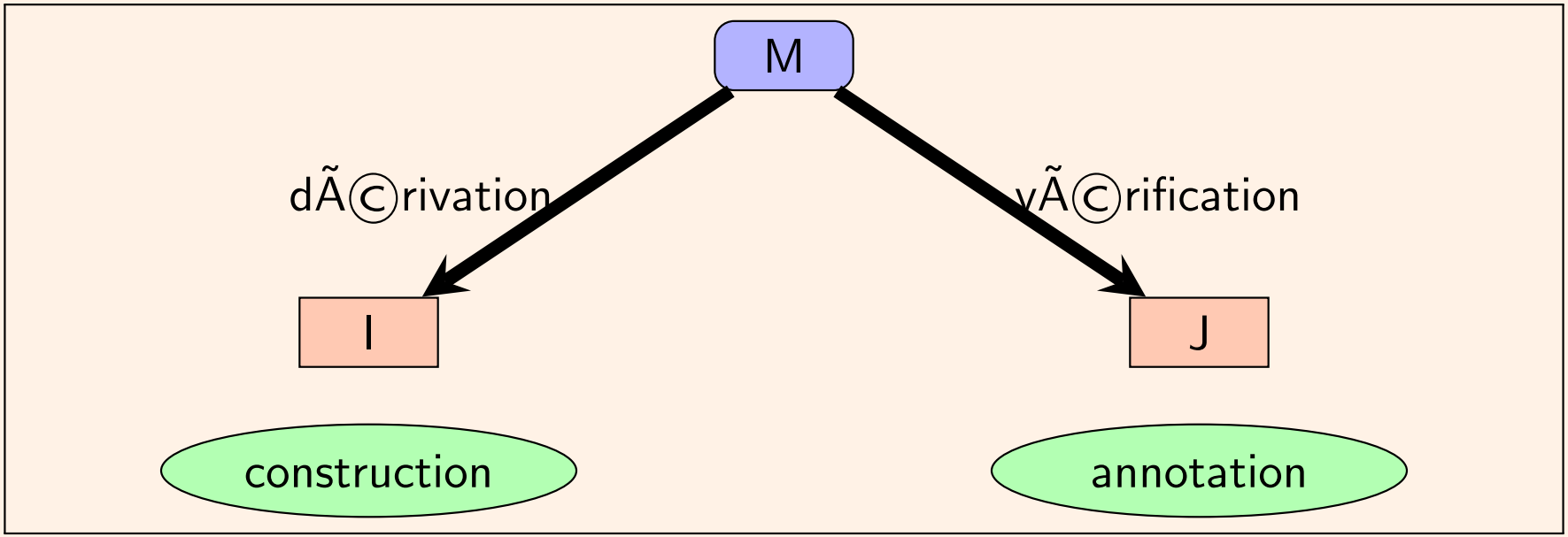
Using the relational induction principle (RI)

- $\textsc{Values} = \textsc{Locations} \times \textsc{Memory}$
- $J(pc_0, v_0, pc, v) \stackrel{\Delta}{=} \exists x_0, x \in \textsc{Values}. I(x_0, x) \wedge x = (pc, v) \wedge x_0 = (pc_0, v_0)$ (deduction)
- $I(x_0, x) \stackrel{\Delta}{=} \exists pc_0, pc \in \textsc{Locations}, v_0, v \in \textsc{Memory}. J(pc_0, v_0, pc, v) \wedge x = (pc, v) \wedge x_0 = (pc_0, v_0)$ (induction)



Using the assertional induction principle (AI)

- $\textsc{Values} = \textsc{Locations} \times \textsc{Memory}$
- $J(pc, v) \triangleq \exists x \in \textsc{Values}. I(x) \wedge x = (pc, v)$ (deduction)
- $I(x) \triangleq \exists pc \in \textsc{Locations}, v \in \textsc{Memory}. J(pc, v) \wedge x = (pc, v)$ (induction)



Alignment with programs

(1) $\forall x_0, x \in \textsc{Values}. Init(x_0) \wedge \textsc{Next}^\star(x_0, x) \Rightarrow A(x)\ (I(x))$
(2) $\forall x_0, x \in \textsc{Values}. Init(x_0) \wedge \textsc{Next}^\star(x_0, x) \Rightarrow R(x_0, x)\ (IR(x))$

Relations and definitions

$x = (\ell, v)$, $x_0 = (\ell_0, v_0)$, $I(x)$, $IR(x_0, x)$ and annotations $P_\ell(v)$, $RP_\ell(v_0, v)$ are linked in this way :

- $I(x) \triangleq \exists \ell, v.(\ell \in \textsc{Locations} \wedge v \in \textsc{Memory} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v) \triangleq \exists x.(x \in \textsc{Values} \wedge x = (\ell, v) \wedge I(x))$
- $IR(x_0, x) \triangleq \exists \ell, v, v_0(\ell \in \textsc{Locations} \wedge v \in \textsc{Memory} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge RP_\ell(v_0, v))$
- $RP_\ell(v_0, v) \triangleq \exists x, x_0.(x, x_0 \in \textsc{Values} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge IR(x_0, x))$

The transformation is based on the transition relation defined for each pair of consecutive control labels, which is expressed very simply by the following relational form :

$$x\ r_{\ell,\ell'}\ x' \triangleq (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell'$$

Transition from one label to another label

- The transition from ℓ to ℓ' is possible when the condition $cond_{\ell,\ell'}(v)$ is true for V and when control is in ℓ ($pc = \ell$).
- When the transition is observed, the variables V are transformed as follows : $v' = f_{\ell,\ell'}(v)$.
- The definition of the transition does not express any assumptions relating to an execution strategy, such as fairness, for example.
- $cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v)$ is an expression in which the expressions $cond_{\ell,\ell'}(v)$ and $= f_{\ell,\ell'}(v)$ raise questions of definition :
  - $DOM(\ell,\ell')(v) \triangleq DEF(cond_{\ell,\ell'}(v))(v) \wedge DEF(f_{\ell,\ell'}(v))$
  - $DEF(E(X))(x)$ means that the expression $E(X)$ is defined for x, the current value of X.
- Some transitions may lead to bugs :
  - $DEF(X+1)(x) \triangleq x+1 \in D$, where D is the coding domain of X ; for example, $D = -2^{31}..2^{31}-1$ for a 32-bit coding scheme.
  - $DEF(T(I+1) < V)(t,x,v) \triangleq i+1 \in dom(t) \wedge v \in D \wedge t(i+1) \in D$

Relationships for program statements

$\ell : P_\ell(v_0, v)$
$V := f_{\ell,\ell'}(V)$
$\ell' : P_{\ell'}(v_0, v)$

$\ell_1 : P_{\ell_1}(v_0, v)$
WHILE $B(V)$ **DO**
  $\ell_2 : P_{\ell_2}(v_0, v)$
  $\ldots$
  $\ell_3 : P_{\ell_3}(v_0, v)$
END
$\ell_4 : P_{\ell_4}(v_0, v)$

Traduction

- $(pc = \ell \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell'$
- $cond_{\ell,\ell'}(v) \stackrel{\Delta}{=} TRUE$

Traduction

$$\begin{cases} pc = \ell_1 \wedge b(v) \wedge v' = v \wedge pc' = \ell_2 \\ pc = \ell_1 \wedge \neg b(v) \wedge v' = v \wedge pc' = \ell_4 \\ pc = \ell_3 \wedge b(v) \wedge v' = v \wedge pc' = \ell_2 \\ pc = \ell_3 \wedge \neg b(v) \wedge v' = v \wedge pc' = \ell_4 \end{cases}$$

Checking the contract : what the method will be

A program P *satisfies* a contract (pre,post) :

- P transforms a variable v starting from an initial value v_0 and producing a final value v_f : $v_0 \xrightarrow{P} v_f$
- v_0 satisfait pre : pre(v_0) and v_f satisfait post : post(v_0, v_f)
- pre$(v_0) \wedge v_0 \xrightarrow{P} v_f \Rightarrow$ post(v_0, v_f)

requires $pre(v_0)$
ensures $post(v_0, v_f)$
variables V

begin
$0 : P_0(v_0, v)$
$\texttt{instruction}_0$
...
$i : P_i(v_0, v)$
...
$\texttt{instruction}_{f-1}$
$f : P_f(v_0, v)$
end

- $pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$
- $pre(v_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$
- conditions on the transitions ℓ, ℓ' to be defined on the basis of the induction principles.

Checking a contract : a simple example

variables U, V
requires $u_0, v_0 \in \mathbb{N}$
ensures $u_f + v_f = u_0 + v_0$

begin
$0 : u = u_0 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N}$
$U := U + 2$
$1 : u = u_0 + 2 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N}$
$V := V - 2$
$2 : u = u_0 + 2 \wedge v = v_0 - 2 \wedge u_0, v_0 \in \mathbb{N}$
end

Checking to do

$$x = (pc, v) et J(\ell_0, v_0, pc, v) \triangleq \left[\begin{array}{l} \wedge\ pc \in \textsc{Locations} \\ \wedge\ v \in \textsc{Memory} \\ \ldots \\ \wedge\ pc = \ell \Rightarrow P_\ell(v_0, v) \\ \ldots \end{array} \right.$$

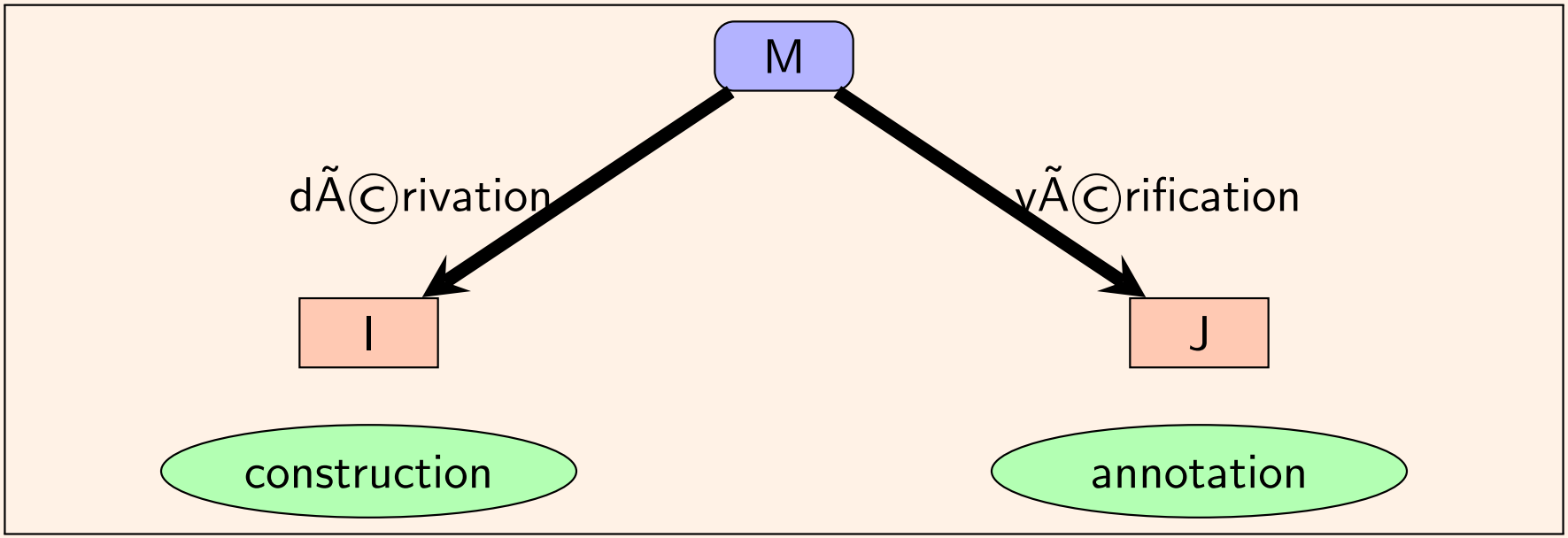
Let $(Th(s,c), x, \textsc{Values}, \textsc{Init}(x), \{r_0, \ldots, r_n\})$ be a relational model for this program. A property $\mathrm{A}(x)$ is a safety property for P if $\forall x_0, x \in \textsc{Locations} \times \textsc{Memory}. Init(x_0) \wedge x_0 \xrightarrow[\textsc{Next}]{} x \Rightarrow A(x)$. We know that this property implies that there exists a state property $\mathrm{I}(x)$ such that the three properties hold, but we apply this verification to J : $\forall x_0, x, x' \in \textsc{Locations} \times \textsc{Memory}$:

$$\left\{ \begin{array}{ll} (1) & \textsc{Init}(x_0) \Rightarrow \mathrm{J}(x_0, x_0) \\ (2) & \mathrm{J}(x_0, x) \Rightarrow \mathrm{A}(x_0, x) \\ (3) & \forall i \in \{0, \ldots, n\} : \mathrm{J}(x_0, x) \wedge x\ r_i\ x' \Rightarrow \mathrm{J}(x_0, x') \end{array} \right.$$

- $\forall i \in \{0, \ldots, n\} : \mathrm{J}(x) \wedge x\ r_i\ x' \Rightarrow \mathrm{J}(x')$ est équivalent à $\mathrm{J}(x) \wedge (\exists i \in \{0, \ldots, n\} : x\ r_i\ x') \Rightarrow \mathrm{J}(x')$

Checking to do

- Application of the correctness to the relational induction principle : if we verify the three properties, then A is a safety property for the model in question (verification).
- If we wish to show that A is a safety property, then we must use the invariant to construct annotations for the model (derivation).



Checking to do : back to the example

- $x = (pc, u, v)$
- $J(0, u_0, v_0, pc, u, v) \triangleq$
$$\left[\begin{array}{l} \wedge\ pc \in \{0,1,2\} \\ \ \wedge\ u, v \in \mathbb{Z} \\ \wedge\ pc = 0 \Rightarrow u = u_0 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N} \\ \wedge\ pc = 1 \Rightarrow u = u_0 + 2 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N} \\ \wedge\ pc = 2 \Rightarrow u = u_0 + 2 \wedge v = v_0 - 2 \wedge u_0, v_0 \in \mathbb{N} \end{array}\right.$$
- $A(0, u_0, v_0, pc, u, v) \triangleq (pc = 2 \Rightarrow u + v = u_0 + v_0 - 2 \wedge u_0, v_0 \in \mathbb{N})$

$\forall pc, u, v, pc', u', v' \in \{0,1,2\} \times \mathbb{Z}$:
$$\left\{\begin{array}{ll} (1) & \textsc{Init}(0, u_0, v_0) \Rightarrow \mathrm{J}(0, u_0, v_0, 0, u_0, v_0) \\ (2) & \mathrm{J}(0, u_0, v_0, pc, u, v) \Rightarrow \mathrm{A}(0, u_0, v_0, pc, u, v) \\ (3) & \forall i \in \{0, \ldots, n\} : \mathrm{J}(0, u_0, v_0, pc, u, v) \wedge x\ r_i\ pc', u', v' \Rightarrow \mathrm{J}(0, u_0, v_0, pc', \end{array}\right.$$

Checking the example

1. $\textsc{Init}(0, u_0, v_0,) \Rightarrow \mathrm{J}(0, u_0, v_0, 0, u_0, v_0)$:
   $pc = 0 \wedge u = u_0 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N} \Rightarrow \mathrm{J}(0, u_0, v_0, 0, u_0, v_0)$:
2. $\mathrm{J}(0, u_0, v_0, pc, u, v) \Rightarrow \mathrm{A}(0, u_0, v_0, pc, u, v)$
   $\mathrm{J}(pc, u, v) \Rightarrow (pc = 2 \Rightarrow u + v = u_0 + v_0 - 2 \wedge u_0, v_0 \in \mathbb{N})$
3. $\forall i \in \{0, \ldots, n\} : \mathrm{J}(0, u_0, v_0, pc, u, v) \wedge x\ r_i\ pc', u', v' \Rightarrow$
   $\mathrm{J}(0, u_0, v_0, pc', u', v')$

$$\left[\begin{array}{l} r01(pc, u, v, pc', u', v') \triangleq pc = 0 \wedge u' = u + 2 \wedge pc' = 1 \wedge v' = v \\ r12(pc, u, v, pc', u', v') \triangleq pc = 1 \wedge v' = v - 2 \wedge pc' = 2 \wedge u' = u \end{array}\right.$$

   - $J(0, u_0, v_0, pc, u, v) \wedge r01(pc, u, v, pc', u', v') \Rightarrow J(pc', u', v')$
   - $J(0, u_0, v_0, pc, u, v) \wedge r12(pc, u, v, pc', u', v') \Rightarrow J(0, u_0, v_0, pc', u', v')$

$J(pc,u,v) \wedge r01(pc,u,v,pc',u',v') \Rightarrow J(pc',u',v')$ (1)

$$
\left(\begin{array}{l}
\wedge\ pc \in \{0,1,2\} \\
\wedge\ u,v \in \mathbb{Z} \\
\wedge\ pc = 0 \Rightarrow u = u_0 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N} \\
\wedge\ pc = 1 \Rightarrow u = u_0 + 2 \wedge v = v_0 \wedge u_0, v_0 \in \mathbb{N} \\
\wedge\ pc = 2 \Rightarrow u = u_0 + 2 \wedge v = v_0 - 2 \wedge u_0, v_0 \in \mathbb{N}
\end{array}\right)
\wedge
\left(\begin{array}{l}
\wedge pc = 0 \\
\wedge u' = u + 2 \\
\wedge pc' = 1 \\
\wedge v' = v
\end{array}\right)
$$

$$
\Rightarrow
$$

$$
\left(\begin{array}{l}
\wedge\ pc' \in \{0,1,2\} \\
\wedge\ u',v' \in \mathbb{Z} \\
\wedge\ pc' = 0 \Rightarrow u' = u_0 \wedge v' = v_0 \wedge u_0, v_0 \in \mathbb{N} \\
\wedge\ pc' = 1 \Rightarrow u' = u_0 + 2 \wedge v' = v_0 \wedge u_0, v_0 \in \mathbb{N} \\
\wedge\ pc' = 2 \Rightarrow u' = u_0 + 2 \wedge v' = v_0 - 2 \wedge u_0, v_0 \in \mathbb{N}
\end{array}\right)
$$

Using the Rodin platform for checking those elements

Initialisation

The relational model $M(P)$ for the annotated program P is therefore defined as follows :
$M(P) \triangleq (Th(s,c),(pc,v),\textsc{Locations} \times$
$\textsc{Memory}, Init(\ell,v), \{r_{\ell,\ell'} | \ell,\ell' \in \textsc{Locations} \wedge \ell \longrightarrow \ell'\})$.
The definition of $Init(x)$ depends on the precondition of P :

$$Init(x) \triangleq .x = (\ell_0, v) \wedge \textbf{pre}(\mathrm{P})(v).$$

Initial conditions

The following two properties are equivalent :

- $\forall x_0 \in \textsc{Values}$: $\mathit{Init}(x_0) \Rightarrow \mathrm{J}(x_0,x_0)$
- $\forall v \in \textsc{Memory}.\textbf{pre}(\mathrm{P})(v) \wedge v = v_0 \Rightarrow P_{\ell_0}(v_0,v)$

Induction step

- Les relations r_i correspondent aux transitions satisfaisant $\ell \longrightarrow \ell'$ et on associe à chaque r_i la relation $r_{\ell,\ell'}$
  - $x\ r_{\ell,\ell'}\ x' \triangleq (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell'$
- $J(x_0, x) \triangleq \exists v_0, \ell, v.(\ell \in \textsc{Locations} \wedge v_0, v \in \textsc{Memory} \wedge x = (\ell, v) \wedge P_\ell(v_0, v))$
- $P_\ell(v_0, v) \triangleq \exists x_0, x.(x_0, x \in \textsc{Values} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$

Induction step

The following two properties are equivalent :

- $\forall i \in \{0, \ldots, n\} : \mathrm{J}(x_0, x) \wedge x\ r_i\ x' \Rightarrow \mathrm{J}(x_0, x')$
- $\forall \ell, \ell' \in \textsc{Locations} : \ell \longrightarrow \ell' \Rightarrow P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$

Induction step (explanation)

- $\mathrm{J}(x_0 n x) \wedge x \; r_{\ell,\ell'} \; x' \Rightarrow \mathrm{J}(x_0, x')$

Induction step (explaination)

- $\mathrm{J}(x_0 n x) \wedge x \; r_{\ell,\ell'} \; x' \Rightarrow \mathrm{J}(x_0, x')$
- $x \; r_{\ell,\ell'} \; x' \stackrel{\Delta}{=} (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell'$

Induction step (explaination)

- $\mathrm{J}(x_0 n x) \wedge x \; r_{\ell,\ell'} \; x' \Rightarrow \mathrm{J}(x_0, x')$
- $x \; r_{\ell,\ell'} \; x' \; \triangleq (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell'$
- $I(x) \triangleq \exists \ell, v.(\ell \in \textsc{Locations} \wedge v \in \textsc{Memory} \wedge x = (\ell, v) \wedge P_\ell(v))$

Induction step (explaination)

- $\mathrm{J}(x_0 n x) \wedge x \; r_{\ell,\ell'} \; x' \Rightarrow \mathrm{J}(x_0, x')$
- $x \; r_{\ell,\ell'} \; x' \; \triangleq (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell'$
- $I(x) \triangleq \exists \ell, v.(\ell \in \textsc{Locations} \wedge v \in \textsc{Memory} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x.(x \in \textsc{Values} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$

Induction step (explaination)

- $\mathrm{J}(x_0 n x) \wedge x \; r_{\ell,\ell'} \; x' \Rightarrow \mathrm{J}(x_0, x')$
- $x \; r_{\ell,\ell'} \; x' \; \triangleq (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell'$
- $I(x) \triangleq \exists \ell, v.(\ell \in \textsc{Locations} \wedge v \in \textsc{Memory} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x.(x \in \textsc{Values} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$
- $\mathrm{J}(x_0 n x) \equiv pc = \ell \wedge P_\ell(v_0, v)$

Induction step (explaination)

- $\mathrm{J}(x_0 n x) \wedge x \; r_{\ell,\ell'} \; x' \Rightarrow \mathrm{J}(x_0, x')$
- $x \; r_{\ell,\ell'} \; x' \; \triangleq (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell'$
- $I(x) \triangleq \exists \ell, v.(\ell \in \textsc{Locations} \wedge v \in \textsc{Memory} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x.(x \in \textsc{Values} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$
- $\mathrm{J}(x_0 n x) \equiv pc = \ell \wedge P_\ell(v_0, v)$
- $\mathrm{J}(x_0, x') \equiv pc = \ell' \wedge P_\ell(v_0, v')$

Induction step (explaination)

- $\mathrm{J}(x_0 n x) \wedge x \; r_{\ell,\ell'} \; x' \Rightarrow \mathrm{J}(x_0, x')$
- $x \; r_{\ell,\ell'} \; x' \; \triangleq (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell'$
- $I(x) \triangleq \exists \ell, v.(\ell \in \textsc{Locations} \wedge v \in \textsc{Memory} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x.(x \in \textsc{Values} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$
- $\mathrm{J}(x_0 n x) \equiv pc = \ell \wedge P_\ell(v_0, v)$
- $\mathrm{J}(x_0, x') \equiv pc = \ell' \wedge P_\ell(v_0, v')$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell' \;\Rightarrow\; pc = \ell' \wedge P_\ell(v_0, v')$

Induction step (explaination)

- $\mathrm{J}(x_0 n x) \wedge x\ r_{\ell,\ell'}\ x' \Rightarrow \mathrm{J}(x_0, x')$
- $x\ r_{\ell,\ell'}\ x' \triangleq (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell'$
- $I(x) \triangleq \exists \ell, v.(\ell \in \textsc{Locations} \wedge v \in \textsc{Memory} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x.(x \in \textsc{Values} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$
- $\mathrm{J}(x_0 n x) \equiv pc = \ell \wedge P_\ell(v_0, v)$
- $\mathrm{J}(x_0, x') \equiv pc = \ell' \wedge P_\ell(v_0, v')$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell' \Rightarrow pc = \ell' \wedge P_\ell(v_0, v')$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell' \Rightarrow pc = \ell'$ (Tautologie)

Induction step (explaination)

- $\mathrm{J}(x_0 n x) \wedge x \; r_{\ell,\ell'} \; x' \Rightarrow \mathrm{J}(x_0, x')$
- $x \; r_{\ell,\ell'} \; x' \; \triangleq (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell'$
- $I(x) \triangleq \exists \ell, v.(\ell \in \textsc{Locations} \wedge v \in \textsc{Memory} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x.(x \in \textsc{Values} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$
- $\mathrm{J}(x_0 n x) \equiv pc = \ell \wedge P_\ell(v_0, v)$
- $\mathrm{J}(x_0, x') \equiv pc = \ell' \wedge P_\ell(v_0, v')$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell' \;\Rightarrow\; pc = \ell' \wedge P_\ell(v_0, v')$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell' \;\Rightarrow\; pc = \ell'$ (Tautologie)
- $pc = \ell \wedge P_\ell(v) \wedge (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell' \;\Rightarrow\; P_{\ell'}(v_0, v')$

Induction step (explaination)

- $\mathrm{J}(x_0 n x) \wedge x\ r_{\ell,\ell'}\ x' \Rightarrow \mathrm{J}(x_0, x')$
- $x\ r_{\ell,\ell'}\ x' \triangleq (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell'$
- $I(x) \triangleq \exists \ell, v.(\ell \in \textsc{Locations} \wedge v \in \textsc{Memory} \wedge x = (\ell, v) \wedge P_\ell(v))$
- $P_\ell(v_0, v) \triangleq \exists x.(x \in \textsc{Values} \wedge x = (\ell, v) \wedge x_0 = (\ell_0, v_0) \wedge J(x_0, x))$
- $\mathrm{J}(x_0 n x) \equiv pc = \ell \wedge P_\ell(v_0, v)$
- $\mathrm{J}(x_0, x') \equiv pc = \ell' \wedge P_\ell(v_0, v')$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell' \ \Rightarrow\ pc = \ell' \wedge P_\ell(v_0, v')$
- $pc = \ell \wedge P_\ell(v_0, v) \wedge (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell' \ \Rightarrow\ pc = \ell'$ (Tautologie)
- $pc = \ell \wedge P_\ell(v) \wedge (pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge \wedge v' = f_{\ell,\ell'}(v) \wedge pc' = \ell' \ \Rightarrow\ P_{\ell'}(v_0, v')$
- $P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$

Conclusion

- $J(x_0, x) \triangleq \exists \ell, v, v_0.(\ell \in \textsc{Locations} \wedge v, v_0 \in \textsc{Memory} \wedge x = (\ell, v) wedge x_0 = (\ell_0, v_0) \wedge P_\ell(v_0, v))$
- $P_\ell(v_0, v) \triangleq \exists x.(x, x_0 \in \textsc{Values} \wedge x = (\ell, v) wedge x_0 = (\ell_0, v_0) \wedge J(x_0), x)$
- $\mathrm{J}(x_0, x) \Rightarrow \mathrm{A}(x_0, x)$
- $\exists \ell, v, v_0.(\ell \in \textsc{Locations} \wedge v, v_0 \in \textsc{Memory} \wedge x = (\ell, v) wedge x_0 = (\ell_0, v_0) \wedge P_\ell(v_0, v)) \Rightarrow \mathrm{A}(x_0, x)$
- $\forall \ell, v, v_0.(\ell \in \textsc{Locations} \wedge v, v_0 \in \textsc{Memory} \wedge x = (\ell, v) wedge x_0 = (\ell_0, v_0) \wedge P_\ell(v_0, v)) \Rightarrow \mathrm{A}(x_0, x)$
- $\forall \ell \in \textsc{Locations}, v, v_0 \in \textsc{Memory}.P_\ell(v_0, v) \Rightarrow \mathrm{A}(\ell_0, v_0, \ell, v)$

Conclusion

The following two properties are equivalent :

- $\mathrm{J}(x_0, x) \Rightarrow \mathrm{A}(x_0, x)$
- $\forall \ell \in \textsc{Locations}, v, v_0 \in \textsc{Memory}.P_\ell(v_0, v) \Rightarrow \mathrm{A}(\ell_0, v_0, \ell, v)$

Translation of the verification conditions

The following verification conditions are equivalent :

- $\forall x_0, x, x' \in \textsc{Locations} \times \textsc{Memory}$:
$$\begin{cases} (1) \quad \textsc{Init}(x_0) \Rightarrow \mathrm{J}(x_0, x_0) \\ (2) \quad \mathrm{J}(x_0, x) \Rightarrow \mathrm{A}(x_0, x) \\ (3) \quad \forall i \in \{0, \ldots, n\} : \mathrm{J}(x_0, x) \wedge x \; r_i \; x' \Rightarrow \mathrm{J}(x_0, x') \end{cases}$$
- $\forall v_0, v, v' \in \textsc{Memory}$:
$$\begin{cases} (1) \quad \textbf{pre}(\mathrm{P})(v_0) \wedge v = v_0 \Rightarrow P_{\ell_0}(v_0, v) \\ (2) \quad \forall \ell \in \textsc{Locations}. P_\ell(v_0, v) \Rightarrow \mathrm{A}(\ell_0, v_0, \ell, v) \\ (3) \quad \forall \ell, \ell' \in \textsc{Locations} : \\ \ell \longrightarrow \ell' \Rightarrow P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v') \end{cases}$$

Method for verification of safety properties

A relational method for proving the correctness of safety properties

- The program is annotated.
- The annotations define an invariant to be verified in accordance with the verification conditions.
- $A(\ell, v)$ is the statement of the safety property to be verified.

Relational method for verifying safety properties

Let $\mathrm{A}(\ell_0, v_0, \ell, v)$ be a property of a program$\mathrm{P}$. Let $\{P_\ell(v_0, v) : \ell \in \textsc{Locations}\}$ be a family of annotations, i.e. a family of properties, for this program. If the following conditions are satisfied : Therefore, $\mathrm{A}(\ell_0, v_0, \ell, v)$ is a safety property for the program P.

Equivalence Floyd/Hoare

Definition

Condition de vérification
L'expression $P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$ où ℓ, ℓ' sont deux étiquettes liées par la relation $\longrightarrow$, est appelée une condition de vérification.

Floyd and Hoare

- $\forall v_0, v, v' \in \textsc{Memory}.\forall \ell, \ell' \in \textsc{Locations}.\ell \longrightarrow \ell' \Rightarrow P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$ is equivalent to $\forall \ell, \ell' \in \textsc{Locations}.\ell \longrightarrow \ell' \Rightarrow \forall v' \in \textsc{Memory}.P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v \mapsto f_{\ell,\ell'}(v))$
- $\forall v_0, v, v' \in \textsc{Memory}.\forall \ell, \ell' \in \textsc{Locations}.\ell \longrightarrow \ell' \Rightarrow P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$ est équivalent à $\forall \ell, \ell' \in \textsc{Locations}.\ell \longrightarrow \ell' \Rightarrow \forall v' \in \textsc{Memory}.\ (\exists v \in \textsc{Memory}.P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v)) \Rightarrow P_{\ell'}(v_0, v')$

Verification condition for the assignment

We can summarise the two possible forms of the assignment as follows :

- $\forall v, v' \in \textsc{Memory}. P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$

$\ell : P_\ell(v_0, v)$
$V := f_{\ell,\ell'}(V)$
$\ell' : P_{\ell'}(v_0, v)$

Verification condition for the assignment

We can summarise the two possible forms of the assignment as follows :

- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge TRUE \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$

$\ell : P_\ell(v_0, v)$
$V := f_{\ell,\ell'}(V)$
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- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge TRUE \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$

$\ell : P_\ell(v_0, v)$
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We can summarise the two possible forms of the assignment as follows :

- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge TRUE \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v, v' \in \text{MEMORY}. P_\ell(v_0, v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v \in \text{MEMORY}. P_\ell(v_0, v) \Rightarrow P_{\ell'}(v_0, v \mapsto f_{\ell,\ell'}(v))$ (Hoare axiomatic system).

$\ell : P_\ell(v_0, v)$
$V := f_{\ell,\ell'}(V)$
$\ell' : P_{\ell'}(v_0, v)$

Verification condition for the assignment

We can summarise the two possible forms of the assignment as follows :

- $\forall v, v' \in \text{MEMORY}.P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v, v' \in \text{MEMORY}.P_\ell(v_0, v) \wedge TRUE \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v, v' \in \text{MEMORY}.P_\ell(v_0, v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')$
- $\forall v \in \text{MEMORY}.P_\ell(v_0, v) \Rightarrow P_{\ell'}(v_0, v \mapsto f_{\ell,\ell'}(v))$ (Hoare axiomatic system).
- $\forall v \in \text{MEMORY}.\ (\exists v' \in \text{MEMORY}.P_\ell(v_0, v) \wedge v' = f_{\ell,\ell'}(v)) \Rightarrow P_{\ell'}(v_0, v')$ corresponds to Floyd assignment rule.

$\ell : P_\ell(v_0, v)$
$V := f_{\ell,\ell'}(V)$
$\ell' : P_{\ell'}(v_0, v)$

Verification condition for the iteration

$\ell_1 : P_{\ell_1}(v_0, v)$
WHILE $B(v)$ **DO**
  $\ell_2 : P_{\ell_2}(v_0, v)$
  $\ldots$
  $\ell_3 : P_{\ell_3}(v_0, v)$
END
$\ell_4 : P_{\ell_4}(v_0, v)$

For the iteration structure, the verification conditions are as follows :

- $P_{\ell_1}(v_0, v) \wedge B(v) \Rightarrow P_{\ell_2}(v_0, v)$
- $P_{\ell_1}(v_0, v) \wedge \neg B(v) \Rightarrow P_{\ell_4}(v_0, v)$
- $P_{\ell_3}(v_0, v) \wedge B(v) \Rightarrow P_{\ell_2}(v_0, v)$
- $P_{\ell_3}(v_0, v) \wedge \neg B(v) \Rightarrow P_{\ell_4}(v_0, v)$

Verification conditions for the conditional

$\ell_1 : P_{\ell_1}(v_0, v)$
IF $B(v)$ **THEN**
  $\ell_2 : P_{\ell_2}(v_0, v)$
  $\ldots$
  $\ell_3 : P_{\ell_3}(v_0, v)$
ELSE
  $m_2 : P_{\ell_2}(v_0, v)$
  $\ldots$
  $m_3 : P_{\ell_3}(v_0, v)$
FI
$\ell_4 : P_{\ell_4}(v_0, v)$

For the conditional statement, the verification cond

- $P_{\ell_1}(v_0, v) \wedge B(v) \Rightarrow P_{\ell_2}(v_0, v)$
- $P_{\ell_3}(v_0, v) \Rightarrow P_{\ell_4}(v_0, v)$
- $P_{\ell_1}(v_0, v) \wedge \neg B(v) \Rightarrow P_{m_2}(v_0, v)$
- $P_{m_3}(v_0, v) \Rightarrow P_{\ell_4}(v_0, v)$

Program correctness properties

Let v be a state variable of P. **pre**$(\mathrm{P})(v)$ is the precondition of P for v ; it characterises the initial values of v. **post**$(\mathrm{P})(v_0, v)$ is the postcondition of P for v ; it characterises the final values of v in relation to the initial value v_0

Example

1. **pre**$(\mathrm{P})(x, y, z)$=$x, y, z \in \mathbb{N}$ et
   post$(\mathrm{P})(x_0, y_0, z_0, x, y, z)$=$z = x_0 * y_0$
2. **pre**$(\mathrm{Q})(x, y, z)$=$x, y, z \in \mathbb{N}$ et
   post$(\mathrm{Q})(x_0, y_0, z_0, x, y, z)$=$z = x_0 + y_0$

$$\forall \underline{x}, \underline{y}, \underline{r}, \underline{q}, \overline{x}, \overline{y}, \overline{r}, \overline{q}.$$
$$\textbf{pre}(\mathrm{P})(\underline{x}, \underline{y}, \underline{r}, \underline{q}) \wedge (\underline{x}, \underline{y}, \underline{r}, \underline{q}) \xrightarrow{\mathrm{P}} (\overline{x}, \overline{y}, \overline{r}, \overline{q})$$
$$\Rightarrow \textbf{post}(\mathrm{P})(\underline{x}, \underline{y}, \underline{r}, \underline{q}, \overline{x}, \overline{y}, \overline{r}, \overline{q})$$

Partial Correctness of a program

Partial correctness aims to establish that a program P is partially correct with respect to its precondition and its postcondition.

- the data specification for P **pre**(P) (v_0)
- the specification of the outputs of P **post**$(\mathrm{P})(\mathrm{v}_0,\mathrm{v})$
- a family of property annotations $\{P_\ell(v_0, v) : \ell \in \textsc{Locations}\}$ for this program.
- a safety property defining the partial correctness $pc = \ell_f \Rightarrow$ **post**$(\mathrm{P})(v_0, v_f)$, where ℓ_f is the label marking the end of the program P

Definition

The program P is partially correct with respect to **pre**$(\mathrm{P})(\mathrm{v}_0)$ and **post**$(\mathrm{P})(\mathrm{v}_0,\mathrm{v})$, if the property $pc = \ell_f \Rightarrow$ **post**$(\mathrm{P})(v_0, v)$ is a safety property for this program.

Partial Correctness of a program

If the following conditions are met :

- $\forall v_0, v \in \text{MEMORY} : \textbf{pre}(\text{P})(v_0) \wedge v = v_0 \Rightarrow P_{\ell_0}(v_0, v)$
- $\forall v_0, v \in \text{MEMORY} : P_{\ell_f}(v_0, v) \Rightarrow \textbf{post}(\text{P})(v_0, v)$
- $\forall \ell, \ell' \in \text{LOCATIONS} : \ell \longrightarrow \ell' : \forall v_0, v, v' \in \text{MEMORY}. (P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v'))$,

So the program P is partially correct with respect to **pre**(P)(v_0) et **post**(P)(v_0,v).

- The partial correctness states that if the program terminates normally, then the postcondition is satisfied by the current variables.
- The semantics of the contract are therefore fairly straightforward to define :

Reformulation of computation

- $\mathsf{pre}(v_0) \wedge v_0 \xrightarrow{\mathsf{P}} v_f \Rightarrow \mathsf{post}(v_0, v_f)$

  (expression of the partial correctness)

- $pc_0 = \ell_0 \wedge \mathsf{pre}(v_0) \wedge (pc_0, v_0) \overset{\textsc{Next}^\star}{\longrightarrow} (pc, v) \wedge pc = \ell_f \Rightarrow \mathsf{post}(v_0, v_f)$

  (big-step semantics and small-step semantics equivalence)

- $pc_0 = \ell_0 \wedge \mathsf{pre}(v_0) \wedge (pc_0, v_0) \overset{\textsc{Next}^\star}{\longrightarrow} (pc, v) \Rightarrow (pc = \ell_f \Rightarrow \mathsf{post}(v_0, v_f))$

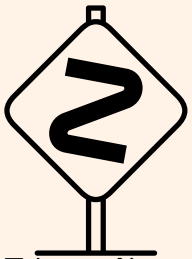
  (implication and conjunction property)

- $Init(x_0) \wedge x_0 \overset{\textsc{Next}^\star}{\longrightarrow} x \Rightarrow \mathsf{PC}(x_0, x)$

$(Init(x_0) \triangleq pc_0 = \ell_0 \wedge \mathsf{pre}(v_0)$

$x_0 \triangleq (\ell_0, v_0)$ and $x \triangleq (pc, v)$

$\mathsf{PC}(x_0 x) \triangleq x_0 = (\ell_0, v_0) \wedge x = (pc, v) \Rightarrow (pc = \ell_f \Rightarrow \mathsf{post}(v_0, v_f))$



Partial correctness is a safety property and the relational method for safety properties is applied.

Checking the contract

A program P *satsifies* a contract (pre,post) :

- P transforms a variable v from an initial value v_0 to a final value v_f : $v_0 \xrightarrow{P} v_f$
- v_0 satisfies pre : pre(v_0) and v_f satisfies post : post(v_0, v_f)
- pre$(v_0) \wedge v_0 \xrightarrow{P} v_f \Rightarrow$ post(v_0, v_f)

requires $pre(v_0)$
ensures $post(v_0, v_f)$
variables V
begin
$0 : P_0(v_0, v)$
$\texttt{statement}_0$
...
$i : P_i(v_0, v)$
...
$\texttt{statement}_{f-1}$
$f : P_f(v_0, v)$
end

- $pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$
- $pre(v_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$
- For any pair of labels ℓ, ℓ' such that $\ell \longrightarrow \ell'$, we verify that, for all values $v, v' \in \text{MEMORY}$

$$\left(\begin{array}{l} \left(\begin{array}{l} P_\ell(v_0, v)) \\ \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \end{array} \right) \\ \Rightarrow P_{\ell'}(v_0, v') \end{array} \right),$$

Checking RTE

- The transition to be executed is the one going from ℓ to ℓ', characterised by the condition or guard $cond_{\ell,\ell'}(v)$ on v and a transformation of the variable v, $v' = f_{\ell,\ell'}(v)$.
- An RTE condition is defined by $\mathbf{DOM}(\ell, \ell')(v)$ for the transition in question. $\mathbf{DOM}(\ell, \ell')(v)$ means that the transition $\ell \longrightarrow \ell'$ is possible and does not lead to an error.
- An error is an arithmetic overflow, a reference to an array element that does not exist, a reference to a null pointer, ...

example

1. The transition corresponds to an assignment of the form $x := x + y$ or $y := x + y$:
   $\mathbf{DOM}(x + y)(x, y) \triangleq \mathbf{DOM}(x)(x, y) \wedge \mathbf{DOM}(y)(x, y) \wedge x + y \in int$
2. The transition corresponds to an assignment of the form $x := x + 1$ ou $y := x + 1$:
   $\mathbf{DOM}(x + 1)(x, y) \triangleq \mathbf{DOM}(x)(x, y) \wedge x + 2 \in int$

Definition RTE

The absence of run time errors is intended to établish that a program P will not produce errors during its execution with regard to its precondition and postcondition.

- the data specification for P **pre**$(\mathrm{P})(\mathrm{v})$
- the result specification for P **post**$(\mathrm{P})(\mathrm{v}_0,\mathrm{v})$
- a family of property annotations $\{P_\ell(v) : \ell \in \textsc{Locations}\}$ for this program.
- a safety property defining the absence of errors at execution :

$$\bigwedge_{\ell \in \textsc{Locations} - \{output\}, n \in \textsc{Locations}, \ell \longrightarrow n} (\mathbf{DOM}(\ell, n)(v))$$

Definition

The program P will not produce any errors during execution with respect to **pre**$(\mathrm{P})(\mathrm{v})$ and **post**$(\mathrm{P})(\mathrm{v}_0,\mathrm{v})$, if the property $\bigwedge_{\ell \in \textsc{Locations} - \{output\}, n \in \textsc{Locations}, \ell \longrightarrow n} (\mathbf{DOM}(\ell, n)(v))$ is a safety property for this program.

Induction principle for RTE

RTE = Run Time Error

If the following conditions are satisfied :

- $\forall v_0, v \in \text{MEMORY} : \textbf{pre}(\text{P})(v_0) \wedge v = v_0 \Rightarrow P_{\ell_0}(v_0, v)$
- $\forall m \in \text{LOCATIONS} - \{\ell_f\}, n \in \text{LOCATIONS}, \forall v_0, v, v' \in \text{MEMORY} :$ $m \longrightarrow n : \textbf{pre}(\text{P})(v_0) \wedge P_m(v_0, v) \Rightarrow \textbf{DOM}(m, n)(v)$
- $\forall \ell, \ell' \in \text{LOCATIONS} : \ell \longrightarrow \ell' : \forall v_0, v, v' \in$ $\text{MEMORY}. (P_\ell(v_0, v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v_0, v')),$

Therefore, the program P will not produce any run-time error with respect to **pre**(P)(v_0) and **post**(P)(v_0,v).

- First, one must verify the partial correction and then reinforce the assertions of the partial correctness with domain conditions.
- One can therefore derive a contract that also incorporates the verification of RTE.

Relational verification method for RTE

A program P *fulfils* a contract (pre,post) :

- P transforms a variable v from an initial value v_0 to a final value v_f : $v_0 \xrightarrow{P} v_f$
- v_0 satisfies pre : $pre(v_0)$ and v_f satisfies post : $post(v_0, v_f)$
- $pre(v_0) \wedge v_0 \xrightarrow{P} v_f \Rightarrow post(v_0, v_f)$
- $\mathbb{D}$ is the RTE domain of V

requires $pre(v_0)$
ensures $post(v_0, v_f)$
variables V
begin
$0 : P_0(v_0, v)$
$\texttt{instruction}_0$
...
$i : P_i(v_0, v)$
...
$\texttt{instruction}_{f-1}$
$f : P_f(v_0, v)$
end

- $pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$
- $pre(x_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$
- For any pair of labels ℓ, ℓ' such that $\ell \longrightarrow \ell'$, we find that, for all values$v, v' \in \textsc{Memory}$
$$\left(\begin{array}{l} \left(\begin{array}{l} P_\ell(v_0, v)) \\ \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \end{array} \right) \\ \Rightarrow P_{\ell'}(v_0, v') \end{array} \right),$$
- $\forall m \in \textsc{Locations} - \{\ell_f\}, n \in \textsc{Locations}, \forall v_0, v, v' \in \textsc{Memory}$: $m \longrightarrow n$:
$pre(v_0) \wedge P_m(v_0, v) \Rightarrow \mathbf{DOM}(m, n)(v)$

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Event B Machine for checking contract of a program

A program P *satisfies* a contract (pre,post) and P transforms a variable v.

Event B Machine for checking contract of a program

A program P *satisfies* a contract (pre,post) and P transforms a variable v.

requires $pre(v_0)$
ensures $post(v_0, v_f)$
variables V

begin
$0 : P_0(v_0, v)$
$\texttt{statement}_0$
...
$i : P_i(v_0, v)$
...
$\texttt{statement}_{f-1}$
$f : P_f(v_0, v)$
end

Event B Machine for checking contract of a program

A program P *satisfies* a contract (pre,post) and P transforms a variable v.

requires $pre(v_0)$
ensures $post(v_0, v_f)$
variables V

begin
$0 : P_0(v_0, v)$
$\texttt{statement}_0$
...
$i : P_i(v_0, v)$
...
$\texttt{statement}_{f-1}$
$f : P_f(v_0, v)$
end

machine M
sees C
variables pc, v
invariants
$I(pc, v)$
theorems
$pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$
$pre(v_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$
INITIALISATION $Init(v)$
...
e(ℓ, ℓ')
...
end

Event B Machine for checking contract of a program

A program P *satisfies* a contract (pre,post) and P transforms a variable v.

requires $pre(v_0)$
ensures $post(v_0, v_f)$
variables V

begin
$0 : P_0(v_0, v)$
$\texttt{statement}_0$
...
$i : P_i(v_0, v)$
...
$\texttt{statement}_{f-1}$
$f : P_f(v_0, v)$
end

machine M
sees C
variables pc, v
invariants
$I(pc, v)$
theorems
$pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$
$pre(v_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$
INITIALISATION $Init(v)$
...
e(ℓ, ℓ')
...
end

What are the invariants, the theorems and the events, such that M is checked if, and only if, the program P fulfils a contract (pre,post)

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Sketch of the construction

- Writing a context C for defining constants as v_0 and properties (axioms) as $pre(v_0$.
- Defining additional constants and predicates for expressing annotations as for instance the domain D of the problem.
- Defining invariants from annotations
- Defining theorems from conditions for checking partial correctness.
- Defining events from the graph of labels.

Defining the context

- A simple algorithm decreases a variable x initially set to a value x_0 of $\mathbb{N}$
- A set L of labels is used for annotating the algorithm.
- Two constants *maxi* and *mini* for expressing the domain of values x_0

variables x
requires $x_0 \in \mathbb{N}$
ensures $x_f = 0$
begin
$\ell_0 : \{ x = x_0 \wedge x_0 \in \mathbb{N}\}$
while $0 < x$ do
  $\ell_1 : \{0 < x \leq x_0 \wedge x_0 \in \mathbb{N}\}$
$x := x - 1;$
$\ell_2 : \{0 \leq x \leq x_0 \wedge x_0 \in \mathbb{N}\}$
od;
$\ell_3 : \{x = 0\}$
end

Defining the context

- A simple algorithm decreases a variable x initially set to a value x_0 of $\mathbb{N}$
- A set L of labels is used for annotating the algorithm.
- Two constants *maxi* and *mini* for expressing the domain of values x_0

variables x
requires $x_0 \in \mathbb{N}$
ensures $x_f = 0$
begin
$\ell_0 : \{ x = x_0 \wedge x_0 \in \mathbb{N}\}$
while $0 < x$ do
$\ell_1 : \{0 < x \leq x_0 \wedge x_0 \in \mathbb{N}\}$
$x := x - 1;$
$\ell_2 : \{0 \leq x \leq x_0 \wedge x_0 \in \mathbb{N}\}$
od;
$\ell_3 : \{x = 0\}$
end

CONTEXT CONTEXT
SETS L
CONSTANTS $x0\ l0\ l1\ l2\ l3\ mini\ maxi$
AXIOMS
$@pre\ x0 \in \mathbb{N}$
$@axm2\ partition(L, \{l0\}, \{l1\}, \{l2\}, \{l3\})$
$@axm3\ mini \in \mathbb{Z} \wedge maxi \in \mathbb{Z}$
$@axm4\ mini < 0 \wedge maxi > 0$
$@axm5\ x0 \leq maxi$
end

From annotations to invariants

$$
\begin{array}{l}
\quad inv_i : pc \in \textsc{Locations} \\
\quad inv_j : v \in Type \\
\cdots \\
\quad inv_k : pc = \ell \Rightarrow P_\ell(v) \\
\quad inv_m : pc = \ell' \Rightarrow P_{\ell'}(v) \\
\cdots \\
\quad th_n : A(pc, v)
\end{array}
$$

- $Type$ is the type of the variables v and is a set of possible values defined in the context C.
- The annotation is giving us for free the conditions satisfied by v when the control is in ℓ, (resp. in ℓ').
- $A(c, v)$ is a safety property that we are supposed to check and the case of Event-B, it is a theorem.

Defining the invariant for our example

- The counter variable is ofte denoted *pc* or *l* or *c*.
- Theorems are stating the conditions of initialistion and finalisation and the RTE conditions.

INVARIANTS

$@inv1\ x \in \mathbb{Z}$

$@inv2\ l \in L$

$@inv3\ l = @xtruc1$

$l0 \Rightarrow x = x0 \wedge x0 \in \mathbb{N}$

$@inv4\ l = l1 \Rightarrow 0 < x \wedge x \leq x0 \wedge x0 \in \mathbb{N}$

$@inv5\ l = l2 \Rightarrow 0 \leq x \wedge x \leq x0 \wedge x0 \in \mathbb{N}$

$@inv6\ l = l3 \Rightarrow x = 0$

$theorem\ @precond\ x = x0 \wedge x0 \in \mathbb{N} \Rightarrow x = x0 \wedge x0 \in \mathbb{N}$

$theorem\ @postcond\ x = 0 \Rightarrow x = 0$

$theorem\ @rte1\ x \in \mathbb{Z}$

$theorem\ @overflow\ mini \leq x \wedge x \leq maxi$

Defining events for transition steps

$$\begin{array}{l}\ell : \{P_\ell(v)\} \\ \quad cond_{\ell,\ell'}(v) \longrightarrow v := f_{\ell,\ell'}(v) \\ \ell' : \{P_{\ell'}(v)\}\end{array}$$

$$\begin{array}{l}\ell_0^1 : \{x = 0\} \\ \quad x := x + 1; \\ \ell_0^1 : \{x = 1\}\end{array}$$

$$\begin{array}{l}e(\ell, \ell') \\ \quad \textbf{WHEN} \\ \quad\quad pc = \ell \\ \quad\quad cond_{\ell,\ell'}(v) \\ \quad \textbf{THEN} \\ \quad\quad pc := \ell' \\ \quad\quad v := f_{\ell,\ell'}(v) \\ \quad \textbf{END}\end{array}$$

- v is the state meory variable or list of memory variables ; v includes the local variables and the results variables.
- pc is a new variable which is modelling the control flow and its type is LOCATIONS.
- $e(\ell, \ell')$ is simulating the computation flow starting from ℓ and moving to ℓ' ; v is updated.

Defining the events for the example

EVENT $INITIALISATION$
then
$@act1\ x := x0$
$@act2\ l := l0$
end
EVENT $el0l1$
where
$@grd1\ l = l0$
$@grd2\ 0 < x$
then
$@act1\ l := l1$
end
EVENT $el0l3$
where
$@grd1\ l = l0$
$@grd2\ \neg(\ 0 < x)$
then
$@act1\ l := l3$
end

EVENT $el1l2$
where
$@grd1\ l = l1$
then
$@act1\ l := l2$
$@act2\ x := x - 1$
end
EVENT $el2l1$
where
$@grd1\ l = l2$
$@grd2\ 0 < x$
then
$@act1\ l := l1$
end
EVENT $el2l3$
where
$@grd1\ l = l2$
$@grd2\ \neg(0 < x)$
then
$@act1\ l := l3$
end

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Verification conditions to check for PC

- $pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$
- $pre(v_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$
- PC (Partial Correctness)
  For any pair of labels ℓ, ℓ'
  such that $\ell \longrightarrow \ell'$,We check that, for all values $v, v' \in \textsc{Memory}$
  $$\left(\begin{array}{l} \left(\begin{array}{l} P_\ell(v_0, v)) \\ \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \end{array} \right) \\ \Rightarrow P_{\ell'}(v_0, v') \end{array} \right),$$
  i

Verification conditions to check for PC

- $pre(v_0) \wedge v = v_0 \Rightarrow P_0(v_0, v)$ **THEOREM**
- $pre(v_0) \wedge P_f(v_0, v) \Rightarrow post(v_0, v)$ **THEOREM**
- PC (Partial Correctness)
  For any event $\mathsf{e}(\ell, \ell')$ with $\ell \longrightarrow \ell'$,
  for all values $v, v' \in \textsc{Memory}$

$$\left(\begin{array}{l} \left(\begin{array}{l} P_\ell(v_0, v)) \\ \wedge BA(\mathsf{e}(\ell, \ell'))(pc, v, pc', v') \end{array} \right) \\ \Rightarrow P_{\ell'}(v_0, v') \end{array} \right),$$

Partial correctness using Event-B models

- Introduction of the definition of events between two successive labels.

$BA(\mathsf{e}(\ell, \ell'))(pc, v, pc', v')$

For each pair of successive labels ℓ, ℓ', the three statements are equivalent :

- $P_\ell(v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v')$
- $I(pc, v) \wedge pc = \ell \wedge cond_{\ell,\ell'}(v) \wedge pc' = \ell' \wedge v' = f_{\ell,\ell'}(v) \Rightarrow (pc' = \ell' \Rightarrow P_{\ell'}(v'))$
- $I(pc, v) \wedge BA(e(\ell, \ell'))(pc, v, pc', v') \wedge pc = \ell \Rightarrow (pc' = \ell' \Rightarrow P_{\ell'}(v'))$

Soundness for PC

Partcial Correctness

Let AA an annotated algorithm with precondition **pre**(AA)(v) and postcondition **post**(AA)(v$_0$,v). Let the context C and the machine M generated from AA using the construction given previously. We assume that ℓ_0 is the first label and ℓ_e is the last label. We add the following properties in the machine M :

- $pc = \ell_0 \wedge$ **pre**$(\mathrm{AA})(v) \Rightarrow P_{\ell_0}(v)$ (condition of initialisation as THEOREM)
- $pc = \ell_e \Rightarrow (P_{\ell_e}(v) \Rightarrow$ **post**$(\mathrm{AA})(v_0, v)$ (condition of safety as THEOREM).

If proof obligations are discharged, then the annotated algorithm AA is partially correct with respect to its pre/post specification.

- The first added theorem is expressing the sound initialisation.
- The second added property is expressing the partial correctness.
- The discharged proof obligations are corresponding to the induction steps.

Checking invariance and safety properties

A relational method for proving the correctness of safety properties

- The contract is stated for variables v namely $v : [pre(v_0, post(v_0, v_f]$
- The program is annotated on the set of labels denoted $\textsc{Locations}$ ($\ell \in \textsc{Locations}$).
- The annotations define an invariant to be verified according to the verification conditions.
- $A(\ell, v)$ is the statement of the safety property to be verified.

A relational method for proving safety properties

Let $\mathrm{A}(\ell_0, v_0, \ell, v)$ be a property of a program P. Let $\{P_\ell(v_0, v) : \ell \in \textsc{Locations}\}$ be a family of annotations, i.e. a family of properties, for this program. If the following conditions are satisfied :
$\forall v, v' \in \textsc{Memory}$:

$$\begin{cases} (1) \quad Pre(v) \Rightarrow P_{\ell_0}(v) \\ (2) \quad \forall \ell \in \textsc{Locations}. P_\ell(v) \Rightarrow \mathrm{A}(\ell, v) \\ (3) \quad \forall \ell, \ell' \in \textsc{Locations} : \\ \ell \longrightarrow \ell' \Rightarrow P_\ell(v) \wedge cond_{\ell,\ell'}(v) \wedge v' = f_{\ell,\ell'}(v) \Rightarrow P_{\ell'}(v') \end{cases}$$

Therefore, $\mathrm{A}(\ell_0, v_0, \ell, v)$ is a safety property for the program P.

Checking using the verification Machine

- Adding the properties (2) $\forall \ell \in \text{LOCATIONS}.P_\ell(v) \Rightarrow \text{A}(\ell, v)$ as THEOREMS
- Szfety properties as RTE (Run Time Error), ADF (Absence of Deadlock Freedom), ...
- Expression of safety properties using the contexte or the theory plugin facilities.

From Floyd to Hoare

From Floyd to Hoare

- $\forall x_f, x_0.\mathsf{pre}(x_0) \wedge x_0 \xrightarrow{\mathsf{P}} x_f \Rightarrow \mathsf{post}(x_0, x_f)$

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- $[\text{if } b(x) \text{ then } S1 \text{ else } S2\ ]P(x) = b(x) \wedge [S1]P(x) \vee \text{ not } b(x)\ [S2]P(x)$

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- Frama-c uses the HOARE logic for defining the verification conditions as R. Leino in DAFNY.
- Questions of termination require the wp calculus ...

Current Section Summary

1. CM3
2. Overview of machines, contexts and proof obligations
3. Proof Obligations for Contexts and Machines
   - PO thm/THM (context)
   - PO th/THM (machine)
   - PO evt/inv/INV
   - PO evt/act/FIS
4. Checking contracts using Event- B on the FIDE Rodin
   - Method for proving invariance properties
   - Verification of contract
   - Definition of machine for checking contract
   - Soundness of the technique
5. Example
6. Conclusion

A short example

s

contract SIMPLE
variables x
requires $x_0 \in \mathbb{N}$
ensures $x_f = 0$
begin
$\ell_0 : \{0 \leq x \leq x_0 \wedge x_0 \in \mathbb{N}\}$
while $0 < x$ **do**
$\ell_1 : \{0 < x \wedge x \leq x_0 \wedge x_0 \in \mathbb{N}\}$
$x := x - 1;$
od
$\ell_2 : \{x = 0\}$end

Event $Init$
THEN
$act1 : x := x0$
$act2 : l := l0$

Event $el0l1$
WHEN
$grd1 : l = l0$
$grd2 : 0 < x$
THEN
$act1 : l := l1$

INVARIANTS
$inv1 : x \in \mathbb{N}$
$inv2 : l \in L$
$inv3 : l = l0 \Rightarrow$
$0 \leq x \wedge x \leq x0 \wedge x0 \in \mathbb{N}$
$inv4 : l = l1 \Rightarrow$
$0 < x \wedge x \leq x0 \wedge x0 \in \mathbb{N}$
$inv5 : l = l2 \Rightarrow x = 0$
$requires : x0 \in \mathbb{N} \wedge x = x0$
$\Rightarrow x = x0 \wedge x0 \in \mathbb{N}$
$ensures : x = 0 \wedge x = x0$
$\Rightarrow x = 0$

Event $el0l2$
WHEN
$grd1 : l = l0$
$grd2 : \neg(0 < x)$
THEN
$act1 : l := l2$

Event $el1l0$
WHEN
$grd1 : l = l1$
THEN
$act1 : l := l0$
$act2 : x := x - 1$

mcfsi0-alg-simple

MACHINE $algorithm$ SEES CONTEXT
VARIABLES $x\ l$
INVARIANTS
$@inv1\ x \in \mathbb{Z}$
$@inv2\ l \in L$
$@inv3\ l = l0 \Rightarrow x = x0 \land x0 \in \mathbb{N}$
$@inv4\ l = l1 \Rightarrow 0 < x \land x \leq x0 \land x0 \in \mathbb{N}$
$@inv5\ l = l2 \Rightarrow 0 \leq x \land x \leq x0 \land x0 \in \mathbb{N}$
$@inv6\ l = l3 \Rightarrow x = 0$
$theorem\ @precond\ x = x0 \land x0 \in \mathbb{N} \Rightarrow x = x0 \land x0 \in \mathbb{N}$
$theorem\ @postcond\ x = 0 \Rightarrow x = 0$
$theorem\ @rte1\ x \in \mathbb{Z}$
$theorem\ @overflow\ mini \leq x \land x \leq maxi$

mcfsi0-alg-simple

EVENT $INITIALISATION$
then
$@act1\ x := x0$
$@act2\ l := l0$
end
EVENT $el0l1$
where
$@grd1\ l = l0$
$@grd2\ 0 < x$
then
$@act1\ l := l1$
end

EVENT $el0l3$
where
$@grd1\ l = l0$
$@grd2\ \neg(\ 0 < x)$
then
$@act1\ l := l3$
end

EVENT $el1l2$
where
$@grd1\ l = l1$
then
$@act1\ l := l2$
$@act2\ x := x - 1$
end
EVENT $el2l1$
where
$@grd1\ l = l2$
$@grd2\ 0 < x$
then
$@act1\ l := l1$
end
EVENT $el2l3$
where
$@grd1\ l = l2$
$@grd2\ \neg(0 < x)$
then
$@act1\ l := l3$
end

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Summary on the technique for verifying safety properties

- Formalizing safety properties as state properties
- Defining the context for expressing the precondition and the postcondition if necessary.
- Using the verification machine, as an engine for cecking safety properties and especially the contract for partial correctness.

- Handling termination by using convergent events and variants.
- Version using the Floyd method contrary to Frama-c.